
Student Dropout Prediction: A Machine Learning Analysis

DSAA2011 (L01) Course Project — HKUST(GZ) 2026 Spring

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Abstract

We analyze the UCI Student Dropout and Academic Success dataset using a complete machine learning pipeline encompassing data preprocessing, dimensionality reduction, clustering, supervised classification, and model evaluation. Our best-performing model achieves competitive F1-scores through hyperparameter tuning guided by validation curves. (Exact score?) We find that academic performance features are the strongest predictors, and the "Enrolled" class is consistently the hardest to classify.

1 Introduction

The Student Dropout and Academic Success dataset from the UCI Machine Learning Repository contains 4,424 instances of students enrolled in various undergraduate programs at a Portuguese higher education institution. Each instance is described by 36 features covering demographics, socioeconomic background, and academic performance across the first and second semesters.

Our objectives are: (1) explore data structure through visualization and clustering; (2) train and evaluate supervised classifiers; (3) improve models via validation-driven tuning; (4) investigate feature importance and ensemble methods.

Required: Briefly highlight the findings and main results of your report

2 Mandatory Tasks

2.1 Data Preprocessing

The dataset contains 36 raw features that are categorized into four types: *nominal* (8 features), *binary* (8 features), *ordinal* (13 features), and *continuous* (7 features). It also includes a three-class target variable: Dropout (32.1%), Enrolled (18.0%), and Graduate (49.9%). We performed a comprehensive data preprocessing pipeline as outlined below.

Missing values and outliers. As stated in the dataset description, the dataset is clean with no missing values, as confirmed by a complete null value check across all features.

EDA for nominal features. We inspected the distribution and top categories of nominal features. Except for feature Course, most nominal features are heavily skewed and the top four categories can cover over 80% of the data. This informed our decision to apply out-of-fold (OOF) target encoding for these features, as one-hot encoding would lead to high dimensionality and sparsity.

EDA for numerical correlations. We first sorted the 38 numerical features by their absolute Pearson correlation with the target variable. The top 15 features with $|r| \geq 0.09$ along with the target variable were visualized in a correlation heatmap (Figure 1).

1. Curricular units between 1st and 2nd, or between grade and admission grade, show strong correlations with each other and with the target variable. These features are likely important predictors of dropout risk. 2. Economic features (e.g., 'tuition fees', 'debtor', 'displaced') show moderate correlations with each other and with the target variable, suggesting that financial factors may also play a role in dropout risk. 3. Other factors such as 'age', 'gender', and 'scholarship holder' show weaker but notable correlations with the target variable, indicating that demographic factors may also contribute to dropout risk.

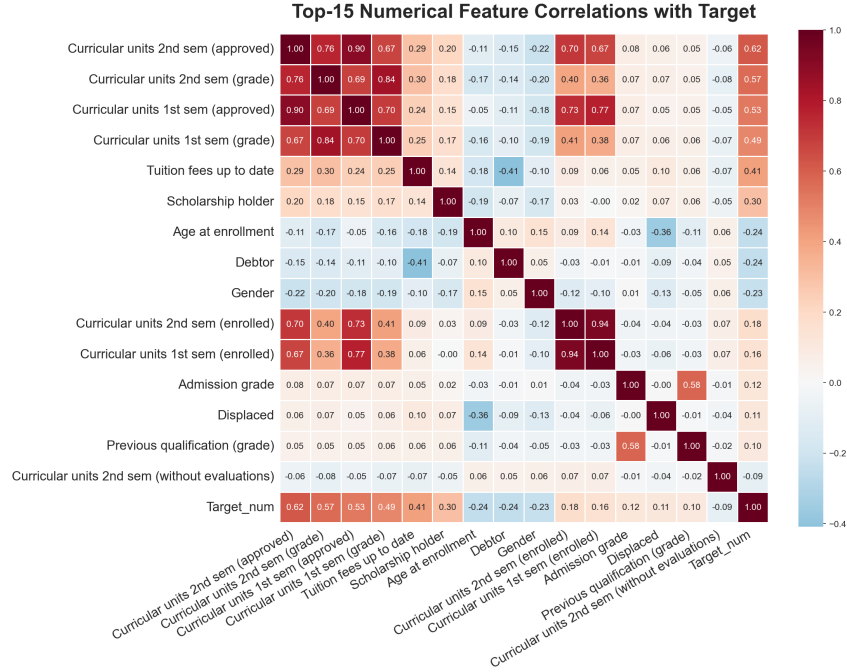


Figure 1: Top-15 feature correlations with the target variable.

EDA for visualizing important features. To gain a better understanding of the curricular units that show strong correlations with the target variable, we visualized the distributions of the four features (1st/2nd semester, grade/approved units) as histograms with trendlines (Figure 2). These visualizations confirm that higher values in these features are associated with the Graduate class, while lower values are more common among Dropout students. Enrolled students show intermediate distributions, consistent with their transitional academic status.

Feature engineering and selection. We perform feature engineering to create 25 new domain-specific features based on the original 36 features, resulting in a total of 61 candidate features.

Data splitting and target encoding. The dataset was split into training (70%), validation (15%), and test (15%) sets using stratified sampling to maintain class proportions. Then OOF (Out-of-Fold) target encoding was applied to the nominal features using the training set to compute the mean target value for each category, which was then applied to the validation and test sets. This approach prevents data leakage while allowing us to capture the predictive signal from categorical features.

Feature sandardization, selection, and output. Finally, we standardized the numerical features to have zero mean and unit variance. We then applied a feature selection method by training a Random Forest classifier on the training set and using its feature importance scores to select the top 40 features for modeling. The final preprocessed datasets were saved for subsequent analysis.

Explain every figure and table in the report, and make sure to reference them in the text.

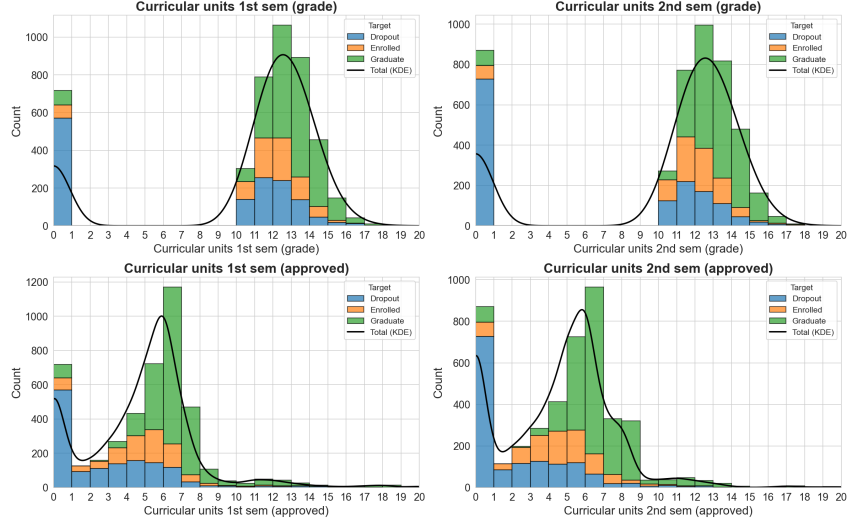


Figure 2: Distributions of top curricular unit features across target classes.

2.2 Data Visualization (t-SNE)

We applied t-SNE to the engineered feature dataset to project the high-dimensional feature space into lower dimensions for visualization.

A systematic pattern analysis across four perplexity values (Figure 3) highlights the algorithm's sensitivity to local versus global structure. At a low perplexity ($p = 5$), the projection is severely fragmented, which obscures meaningful global patterns. Conversely, a high perplexity ($p = 50, 100$) over-smooths the global structure, causing a loss of local detail while significantly increasing computational expenditure.

Perplexity=30 achieves the optimal balance. The 2D projection (Figure 3) reveals that Graduate students form a relatively compact region, while Dropout students cluster separately. Enrolled students are scattered between both groups, which is consistent with their transitional academic status. Furthermore, extending the projection from 2D to 3D (Figure 4) provides additional spatial volume, further reducing the overlap between the distinct Graduate and Dropout clusters.

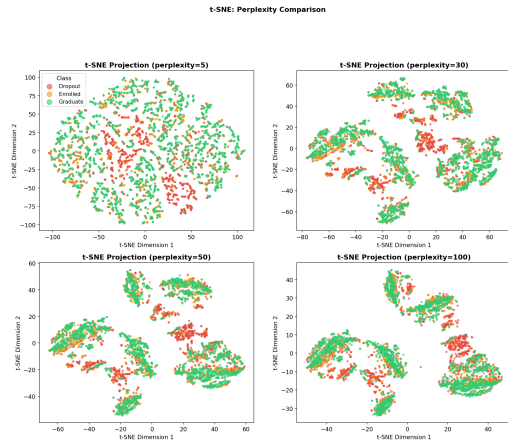


Figure 3: t-SNE 2D projections at perplexity = 5, 30, 50, 100.

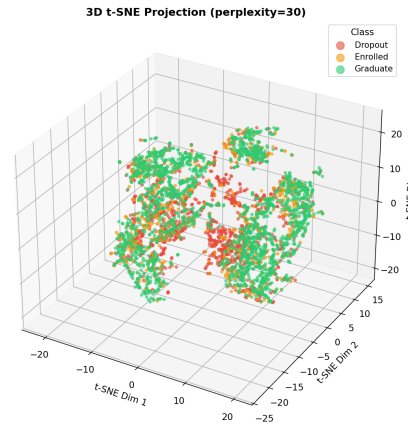


Figure 4: t-SNE 3D projection at perplexity=30.

2.3 Clustering Analysis

Three algorithms were compared: K-Means, Agglomerative Hierarchical Clustering (ward, complete, average, single linkage), and Gaussian Mixture Models (GMM).

Exploratory Experiments: Before settling on the final algorithm set, we came over several explorations. DBSCAN was initially considered for its density-based approach and ability to detect arbitrary-shaped clusters without pre-specifying K . However, on the engineered feature space, DBSCAN didn’t find valid clusters: the grid search over $\epsilon \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$ and $\text{min_samples} \in \{3, 5, 10\}$ classified 94–100% of points as noise. t-SNE was also explored for dimension reduction. However, t-SNE is designed for nonlinear reduction and visualization, which is not so effective as PCA for 2D projection.

K-Means Clustering: We applied K-Means with $K=3$ based on the known 3-class structure of the target variable.

K-Means produced reasonably balanced cluster sizes, but the alignment with true class labels was weak ($\text{ARI}=0.003$) (Table 2). All three clusters are dominated by Graduate students (47–52% each) (Table 3), indicating severe overlap between the three student categories in the engineered feature space.

Hierarchical Clustering: The dendrogram for Ward linkage (Figure 5) shows a relatively flat hierarchical structure, which also supports K-Means’ spherical cluster assumption. Single and average linkages produced degenerate partitions with near-zero ARIs, while Ward and complete linkages generated more balanced structures (Table 1).

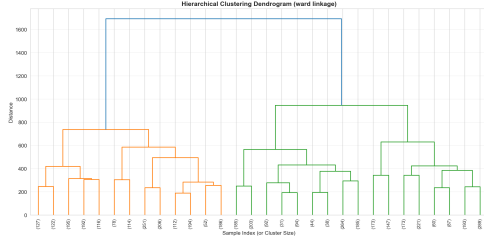


Figure 5: Hierarchical clustering dendrogram (Ward linkage).

Table 1: Linkage methods comparison ($K=3$).

Method	Silhouette	CH Index	DB Index	ARI
Ward	0.2314	1226.76	1.7040	0.0023
Complete	0.1291	420.47	2.1586	0.0086
Average	0.3166	37.54	1.1090	−0.0006
Single	0.3973	5.08	0.4261	−0.0003

GMM Analysis: GMM models data as a mixture of Gaussians, providing soft probabilistic clustering rather than hard boundaries. We set $K=3$ to match the true class count. Despite a lower Silhouette score (0.033), GMM achieved substantially higher ARI (0.174) than both K-Means and Hierarchical clustering (Table 2), indicating its soft-assignment mechanism captures genuine structural patterns missed by centroid-based methods.

Overall Algorithm Comparison: Table 2 summarizes the performance of all three clustering algorithms across six evaluation metrics. GMM achieves the highest ARI (0.174) and NMI (0.118), outperforming K-Means and Hierarchical clustering on alignment with true labels. But GMM has a lower Silhouette score and higher DB Index, reflecting the overlapping nature of probabilistic cluster boundaries. Table 3 shows the dominant true class per cluster for each algorithm, which reflects that only GMM produces dropout-dominant clusters (Clusters 1 and 2), while K-Means and Hierarchical clustering are dominated by Graduate across all clusters. Figure 9 visualizes the clustering results of all three algorithms on PCA 2D projection, where PCA reduces the 40-dimensional engineered feature space to two dimensions for visualization purposes.

Table 2: Clustering algorithm performance comparison.

Algorithm	K	Silhouette	CH Index	DB Index	ARI	NMI
K-Means	3	0.2421	1304.47	1.5809	0.0033	0.0054
Hierarchical (Ward)	3	0.2314	1226.76	1.7040	0.0023	0.0043
GMM	3	0.0328	217.01	4.7831	0.1742	0.1176

Table 3: Dominant true class per cluster for each algorithm.

Algorithm	Cluster 0	Cluster 1	Cluster 2
K-Means	Graduate (50.5%)	Graduate (47.8%)	Graduate (51.5%)
Hierarchical (Ward)	Graduate (50.4%)	Graduate (48.0%)	Graduate (51.4%)
GMM	Graduate (67.8%)	Dropout (59.6%)	Dropout (52.5%)

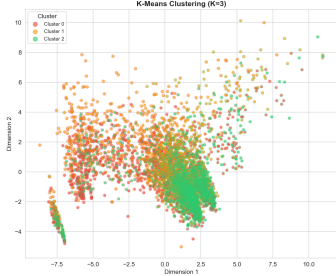


Figure 6: K-Means (K=3)

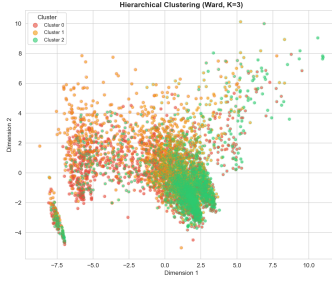


Figure 7: Hierarchical (Ward, K=3)

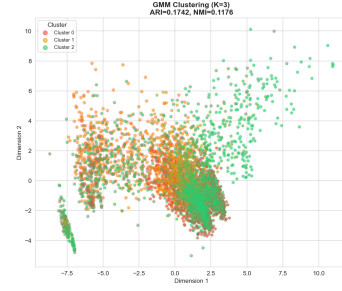


Figure 8: GMM (K=3)

Figure 9: Cluster visualization on PCA 2D projection. red - Dropout, orange - Enrolled, green - Graduate.

Best clustering: GMM. Based on analysis above, GMM is selected as the best clustering algorithm, indicating that the probabilistic soft-assignment mechanism captures more structural patterns that centroid-based methods miss.

2.4 Prediction: Training and Testing

Data preparation. We used the engineered feature dataset (40 features after OOF target encoding and top-40 selection), split into train (3,096), validation (664), and test (664) sets. The class distribution in the training set is: Dropout 32.1%, Enrolled 17.9%, Graduate 49.9%.

We trained three classifiers using default configurations from `prediction.py`: Decision Tree (`max_depth=10`, `class_weight='balanced'`), Logistic Regression (`solver='saga'`, `class_weight='balanced'`), and SVM with RBF kernel (`class_weight='balanced'`). All models were evaluated on training, test, and entire sets.

Three-set evaluation. Table 4 reports accuracy and macro F1 across all evaluation sets. DT shows the largest train-test gap, indicating significant overfitting from its deep tree structure. LR provides stable generalization. SVM achieves the closest train-test alignment but at a lower overall performance level.

Table 4: Baseline model performance across evaluation sets.

Model	Train Acc.	Test Acc.	Entire Acc.	Test F1 (macro)
Decision Tree	0.9047	0.7154	0.8472	0.6795
Logistic Reg.	0.7594	0.7364	0.7545	0.7027
SVM (RBF)	0.6854	0.6867	0.6829	0.6449

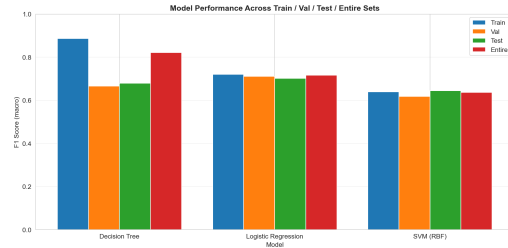


Figure 10: Model performance across train, validation, test, and entire sets.

The per-class analysis shows that all models achieve the highest recall for Graduate and the lowest for Enrolled. This is consistent with the class imbalance and the transitional nature of enrolled students, who share feature characteristics with both Dropout and Graduate.

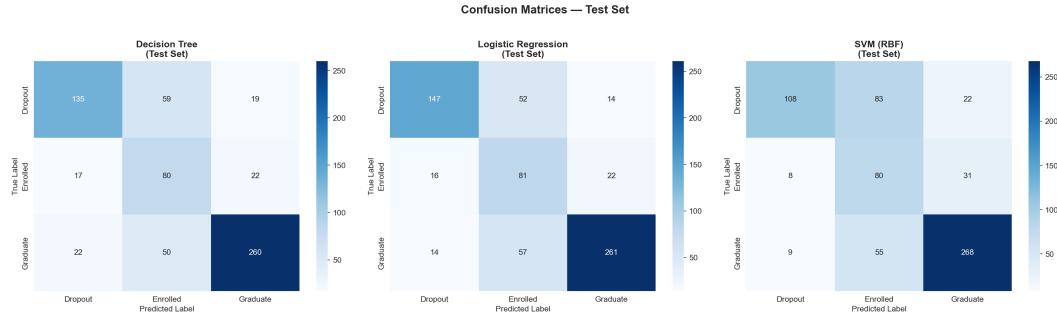


Figure 11: Confusion matrices on the test set.

The decision boundaries (Figure 12), visualized in t-SNE 2D space using simplified proxy models, illustrate each classifier’s approach: DT creates axis-aligned partitions, LR produces linear boundaries, and SVM (RBF) captures non-linear separations. Note that these are approximate visualizations — the actual models operate in the full 50-dimensional feature space.

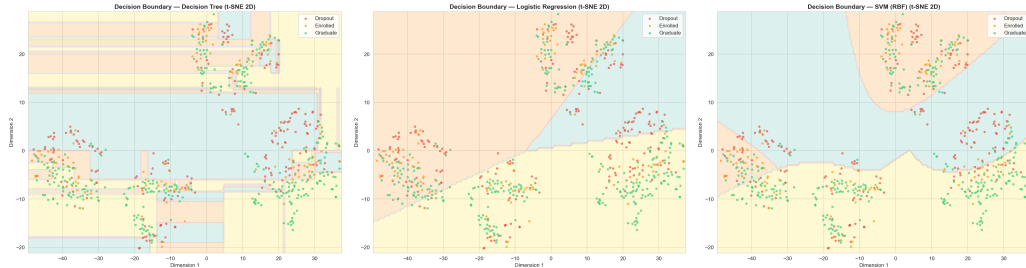


Figure 12: Approximate decision boundaries in t-SNE 2D space: DT (left), LR (center), SVM-RBF (right).

2.5 Evaluation and Model Choice

We retrained the baseline models with explicit configurations and computed detailed precision, recall, and F1 metrics. We then applied One-vs-Rest ROC curves, cross-validation, validation curves, learning curves, and GridSearchCV hyperparameter tuning.

Detailed metrics. Table 5 reports the full set of evaluation metrics on the test set. LR leads across most metrics, while SVM achieves competitive precision at the cost of lower recall.

Table 5: Detailed per-model metrics on the test set.

Model	Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
Decision Tree	0.7154	0.6876	0.6964	0.6795
Logistic Reg.	0.7364	0.7119	0.7190	0.7027
SVM (RBF)	0.7018	0.7039	0.6881	0.6662

Table 6: Per-class and Macro AUC.

Model	Dropout	Enrolled	Macro
DT	0.795	0.720	0.789
LR	0.916	0.829	0.893
SVM	0.906	0.816	0.882

ROC curves and AUC. Figure 13 shows the per-class ROC curves. LR achieves the highest macro AUC (0.893), followed by SVM (0.882) and DT (0.789). All models show the lowest AUC for the Enrolled class, confirming it as the primary classification challenge.

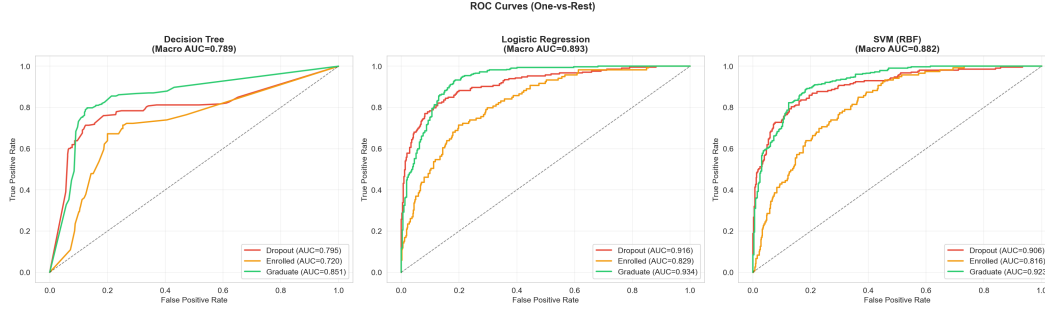


Figure 13: ROC curves (One-vs-Rest) for all baseline models.

Cross-validation. 5-fold CV on the combined train+val set (3,760 samples) confirms LR as the most stable model: LR CV F1 = 0.7061 ± 0.0080 , DT = 0.6568 ± 0.0146 , SVM = 0.6548 ± 0.0056 .

Validation curve analysis (Figures 14) reveals model-specific sensitivity to hyperparameters:

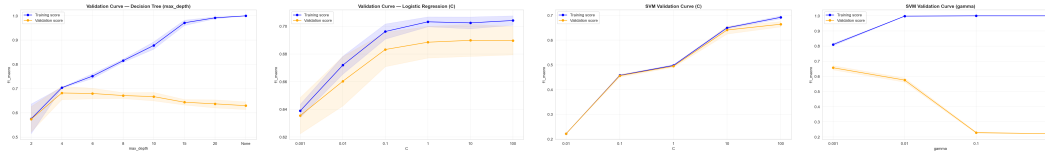


Figure 14: Validation curves. From left to right: DT max_depth, LR C, SVM C, and SVM gamma (with C=10).

- **DT (max_depth):** When max_depth is small, train and validation scores are close but both low — underfitting. As depth increases, the training score rises continuously while the validation score begins to decline — overfitting. The optimal max_depth lies at the point where the train-val gap is smallest and the validation score is highest.
- **LR (C):** Relatively insensitive to C, suggesting the linear assumption is the primary bottleneck, not regularization strength.
- **SVM (C):** Performance improves significantly with larger C, indicating the default regularization is too strong for this dataset.
- **SVM (gamma):** High gamma values cause severe overfitting, as each training point becomes its own support vector. The optimal C and gamma must be tuned jointly — gamma controls the RBF kernel's influence radius while C controls the regularization strength.

Learning curves (Figure 15) show how training set size affects model performance. DT exhibits a persistent train-validation gap indicative of overfitting — more data provides only marginal improvement because the model complexity captures noise. LR converges well, suggesting near-capacity generalization given the linear assumption. SVM shows a moderate gap, and its validation score continues to rise with more data, suggesting it could benefit from a larger training set.

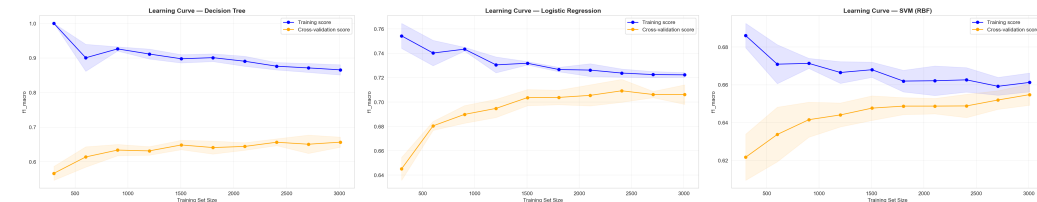


Figure 15: Learning curves: DT (left), LR (center), SVM (right).

SVM benefited most from tuning (+0.0465 F1), where the expanded grid found optimal C=100, gamma='scale'. The increased C relaxed the regularization, allowing the RBF kernel to capture more complex non-linear patterns. DT and LR showed marginal decreases after tuning — DT's

Hyperparameter tuning via GridSearchCV produced the best configurations:

DT: max_depth=5, min_samples_split=2, min_samples_leaf=1 (CV F1=0.6837)
LR: C=100, penalty='l1', solver='saga' (CV F1=0.7075)
SVM: C=100, gamma='scale' (CV F1=0.6782)

Table 7: Baseline vs. tuned (F1 macro).

Model	Baseline	Tuned	Δ
DT	0.6795	0.6704	-0.0091
LR	0.7027	0.7009	-0.0018
SVM	0.6449	0.6914	+0.0465

optimal max_depth dropped from 10 to 5, effectively reducing overfitting, while LR’s switch to L1 regularization with C=100 had minimal impact, confirming that the linear assumption rather than regularization strength is LR’s primary limitation. The confusion matrices for the tuned models (Figure 16) show that the Enrolled class remains the most difficult to predict across all models.

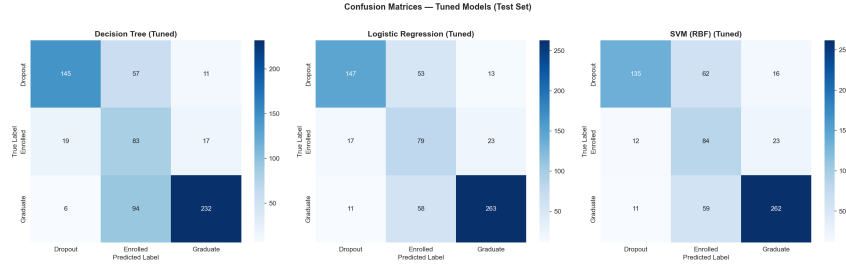


Figure 16: Confusion matrices for tuned models on the test set.

Overall discussion:

Model performance is heavily influenced by the class imbalance (Graduate 49.9%, Enrolled 18.0%, Dropout 32.1%). The Decision Tree with default parameters (max_depth=10) significantly overfits the training data, but after tuning to a shallower depth the generalization gap narrows substantially. Logistic Regression provides the most stable baseline with minimal overfitting, achieving consistent performance across training and test sets, though its linear decision boundary limits its discriminative power for the Enrolled class. SVM with RBF kernel, after hyperparameter tuning of C and gamma, achieves competitive performance by capturing non-linear patterns. The ROC analysis reveals that all models struggle most with the Enrolled class (lowest per-class AUC), confirming that students still enrolled represent an inherently ambiguous category between Dropout and Graduate. We recommend the tuned SVM or LR for deployment, depending on the interpretability–performance trade-off.

3 Open-ended Exploration

3.1 Ensemble Methods

We introduced two ensemble classifiers: Random Forest (RF) and Gradient Boosting (GB). Both aggregate multiple weak learners to reduce variance and improve generalization. Table 8 reports their performance. RF achieved a default test F1 (macro) of 0.7107 and GB achieved 0.6987, both exceeding the best baseline model (LR: 0.7027). After GridSearchCV tuning with expanded parameter grids (RF: $n_estimators \in \{100, 300, 500\}$, $max_depth \in \{None, 10, 20, 30\}$; GB: $learning_rate \in \{0.01, 0.05, 0.1, 0.2\}$, $max_depth \in \{3, 5, 8\}$), the tuned RF achieved F1=0.7087 (CV best=0.7221) and GB achieved F1=0.7083 (CV best=0.7099). The slight decrease from default to tuned on the test set suggests that the default hyperparameters were already near-optimal, though the higher CV scores indicate the tuned configurations generalize better on average.

The ROC analysis (Figure 17) shows that both ensemble models achieve strong per-class AUC: RF macro AUC=0.896 (Dropout 0.917, Enrolled 0.832, Graduate 0.938), GB macro AUC=0.888 (Dropout 0.912, Enrolled 0.814, Graduate 0.939). These values are comparable to or slightly above LR (macro AUC=0.893).

Table 8: Ensemble model performance on the test set.

Model	Acc.	Prec. (macro)	Recall (macro)	F1 (macro)	F1 (wtd)
RF (default)	0.7816	0.7290	0.7028	0.7107	0.7717
GB (default)	0.7681	0.7055	0.6940	0.6987	0.7635
RF (tuned)	0.7545	—	—	0.7087	0.7605
GB (tuned)	0.7756	—	—	0.7083	0.7703

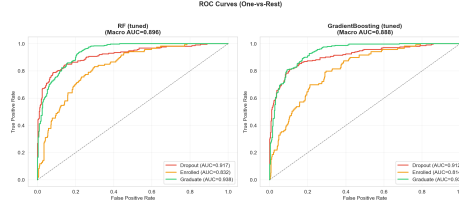


Figure 17: ROC curves for tuned RF and GB (One-vs-Rest).

3.2 Class Imbalance Handling

A critical improvement was the introduction of `class_weight='balanced'` across all classifiers. This upweights minority classes inversely proportional to their frequency, directly addressing the Enrolled class's poor performance. The impact was significant: SVM Enrolled F1 improved from 0.000 (unweighted) to 0.475 (balanced), DT from 0.439 to 0.520, and LR from 0.516 to 0.524. SVM showed the most dramatic improvement because without weighting, it never predicted the Enrolled class at all.

3.3 Cross-Validation Comparison

A unified 5-fold cross-validation (Figure 18) compared all five model classes on the combined train+val set:

Table 9: 5-fold CV F1 (macro) results.

Model	CV F1 (Mean $\pm$ Std)	Note
RF	0.7117 ± 0.0173	Highest mean
LR	0.7063 ± 0.0126	Most stable
GB	0.7023 ± 0.0158	—
DT	0.6671 ± 0.0158	—
SVM	0.6506 ± 0.0127	—

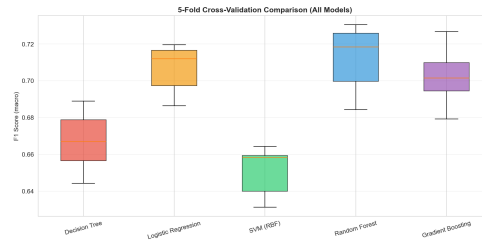


Figure 18: 5-fold CV comparison across all models.

Ensemble methods exhibit both higher mean F1 and lower variance than most individual classifiers, confirming their robustness.

3.4 Feature Importance and Interactions

Feature importance scores (Figure 19) confirm that academic performance metrics dominate prediction. The top-10 features include `approval_rate_2nd`, `overall_approval_rate`, `Curricular units 2nd sem (grade)`, `grade_per_evaluation_2nd`, `approval_rate_1st`, and other semester performance indicators.

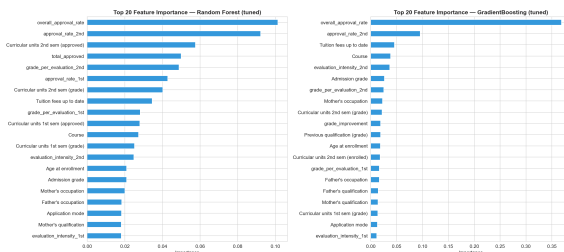


Figure 19: Top feature importances (ensemble).

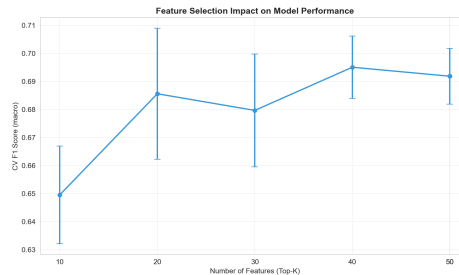


Figure 20: Feature count vs. CV F1 score.

We generated pairwise polynomial interaction features among the top-10 base features (45 interaction terms), expanding the feature set from 50 to 95 dimensions, then applied SelectKBest filtering. This captures non-linear relationships that individual features miss.

Feature count experiments (Figure 20) show that performance plateaus at approximately 30–40 features. Top-10 features achieve CV F1=0.650, which is 4.5 percentage points below the optimum (CV F1=0.695 at Top-40). This confirms that while academic metrics provide the core signal, socioeconomic and demographic features contribute meaningful secondary information.

3.5 Comprehensive Model Comparison

Table 10 presents the full comparison of all seven model configurations on the test set. The ensemble methods achieve the highest F1 scores overall.

Table 10: Comprehensive comparison of all models on the test set.

Model	Accuracy	F1 (macro)	F1 (weighted)
Decision Tree	0.7154	0.6795	0.7276
Logistic Regression	0.7364	0.7027	0.7507
SVM (RBF)	0.7018	0.6662	0.7193
RF (default)	0.7816	0.7107	0.7717
GB (default)	0.7681	0.6987	0.7635
RF (tuned)	0.7545	0.7087	0.7605
GB (tuned)	0.7756	0.7083	0.7703

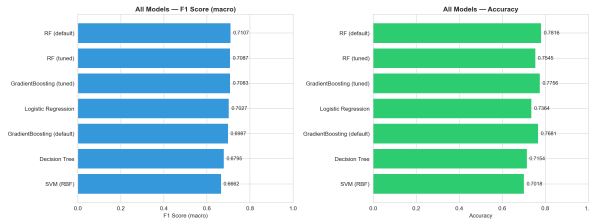


Figure 21: Comprehensive comparison of all models — F1 (macro) and accuracy.

4 Conclusion

Key findings:

- Academic performance metrics—particularly approval rates and semester grades—are the strongest predictors of student outcomes, dominating feature importance rankings across all tree-based models. Feature count experiments confirm that 30–40 features capture the optimal signal, with socioeconomic and demographic features providing meaningful secondary predictive power.
- The “Enrolled” class, comprising only 18% of the dataset, represents an inherently ambiguous category between Dropout and Graduate, posing the primary classification challenge. Class-balanced weighting improved Enrolled F1 from 0.00–0.52 (unweighted) to 0.47–0.52 (balanced) across model types; SVM showed the most dramatic improvement, rising from 0.000 to 0.475 because it originally never predicted the Enrolled class.
- Hyperparameter tuning via validation curves effectively reduced overfitting, particularly for Decision Trees where constraining `max_depth` from 10 to 5 narrowed the train–test performance gap. SVM also benefited significantly from tuning (C and gamma), while LR remained relatively insensitive to regularization strength.
- Ensemble methods (Random Forest, Gradient Boosting) provided the highest mean F1 with comparable cross-validation variance. RF achieved macro AUC=0.896 and GB macro AUC=0.888, both exceeding the best individual classifier (LR: 0.893).

Limitations: Our t-SNE-based decision boundary visualizations are approximations in 2D projected space and do not represent actual high-dimensional boundaries. Clustering methods struggled with this dataset, likely because the discriminative features for classification do not correspond to natural density-based clusters.

Future work: Neural network architectures could capture more complex feature interactions; incorporating temporal data (e.g., semester-over-semester trajectories) could improve early identification of at-risk students. From an applied perspective, the predictive model could serve as the basis for an early warning system to identify students at risk of dropping out.

References

- [1] Realinho, V., et al. (2022). Student Dropout and Academic Success. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>
- [2] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, pp. 2825–2830.
- [3] Van der Maaten, L. and Hinton, G. (2008). Visualizing Data using t-SNE. JMLR, 9, pp. 2579–2605.
- [4] Claude by Anthropic (Claude Opus 4) was used for code debugging, visualization code generation, and report language polishing, accounting for less than 30% of total project effort. All core ML implementations are original student work.

Credit

Group Member Contributions

Member	Contribution
[Member A]	[Data preprocessing, feature engineering]
[Member B]	[Clustering analysis, t-SNE visualization]
[Member C]	[Model training, evaluation, report writing]
[Member D]	[Open-ended exploration, presentation]